Sampling without Replacement, and Variance Estimation

April 28, 2026

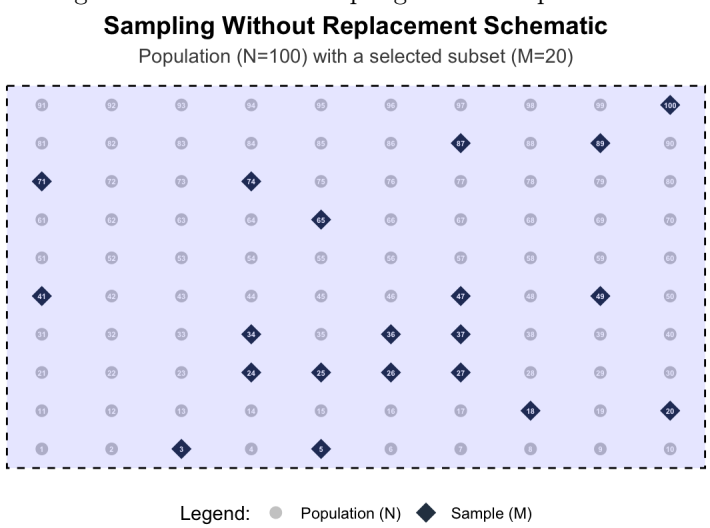
Mean Estimation by Sampling without Replacement

Given a finite population $\{x_i : i \in [N]\}$, consider estimating the mean

$$\mu := \frac{1}{N} \sum_{i \in [N]} x_i$$

by sampling a subset $\mathcal{S}$ of $M \leq N$ of elements in the population *without* replacement, and reporting their sample mean.

Figure 1: Illustration sampling without replacement.



It is known that this estimator is unbiased for μ. Write also

$$\begin{aligned}\sigma^2 &= \frac{1}{N-1} \sum_{i \in [N]} (x_i - \mu)^2 \\ &= \frac{N}{N-1} \cdot \left\{ \left(\frac{1}{N} \sum_{i \in [N]} x_i^2 \right) - \left(\frac{1}{N} \sum_{i \in [N]} x_i \right)^2 \right\}.\end{aligned}$$

Variance Formula

What is the variance of this estimator? Moreover, how can we efficiently compute an estimator for this variance, using only the same set of samples?

Here is one solution: introduce the variable

$$z_i = \mathbf{I}\left[i \in \mathcal{S}\right]$$

so that the proposed estimator can be written as

$$\hat{\mu} = \frac{1}{M} \sum_{i \in [N]} z_i \cdot x_i.$$

Given this representation, compute that

$$\mathbf{E}\left[\hat{\mu}^2\right] = \frac{1}{M^2} \mathbf{E}\left[\sum_{i \in [N]} \sum_{j \in [N]} z_i \cdot z_j \cdot x_i \cdot x_j\right].$$

Now, for $i \neq j$, it holds that

$$\mathbf{E}\left[z_i \cdot z_j\right] = \mathbf{P}\left(x_i, x_j \text{ both included in the sample}\right) = \frac{M \cdot (M-1)}{N \cdot (N-1)}.$$

Similarly, along the diagonal, one can compute that $\mathbf{E}\left[z_i^2\right] = \mathbf{E}\left[z_i\right] = \frac{M}{N}$. Summarising the two cases, we can write that

$$\begin{aligned}
\mathbf{E}\left[z_i \cdot z_j\right] &= \frac{M \cdot (M-1)}{N \cdot (N-1)} + \delta_{i,j} \cdot \left(\frac{M}{N} - \frac{M \cdot (M-1)}{N \cdot (N-1)}\right) \\
&= \frac{M \cdot (M-1)}{N \cdot (N-1)} + \delta_{i,j} \cdot \left(\frac{M \cdot (N-M)}{N \cdot (N-1)}\right),
\end{aligned}$$

where $\delta_{i,j}$ is the Kronecker delta (equal to 1 if $i = j$ and 0 otherwise).

As such, it follows that

$$\begin{aligned}
\mathbf{E}\left[\hat{\mu}^2\right] &= \frac{1}{M^2} \sum_{i \in [N]} \sum_{j \in [N]} \left\{\frac{M \cdot (M-1)}{N \cdot (N-1)} + \delta_{i,j} \cdot \left(\frac{M \cdot (N-M)}{N \cdot (N-1)}\right)\right\} \cdot x_i \cdot x_j \\
&= \frac{M-1}{M} \cdot \frac{N}{N-1} \cdot \mu^2 + \frac{N}{M^2} \cdot \frac{M \cdot (N-M)}{N \cdot (N-1)} \cdot \left(\frac{1}{N} \sum_{i \in [N]} x_i^2\right) \\
&= \mu^2 + \frac{N-M}{M \cdot (N-1)} \cdot \left\{\left(\frac{1}{N} \sum_{i \in [N]} x_i^2\right) - \left(\frac{1}{N} \sum_{i \in [N]} x_i\right)^2\right\},
\end{aligned}$$

i.e.,

$$\mathbf{Var}\left(\hat{\mu}\right) = \frac{N-M}{M \cdot N} \cdot \sigma^2.$$

Variance Estimation

How then do we estimate this quantity? Note first that

$$\begin{aligned}
\sigma^2 &= \frac{1}{N-1} \sum_{i \in [N]} \left(x_i - \mu\right)^2 \\
&= \frac{1}{2 \cdot N \cdot (N-1)} \sum_{i \in [N]} \sum_{j \in [N]} \left(x_i - x_j\right)^2.
\end{aligned}$$

On the other hand, for fixed i, j, it holds that

$$\begin{aligned}
\mathbf{E}\left[z_i \cdot z_j \cdot \left(x_i - x_j\right)^2\right] &= \mathbf{E}\left[z_i \cdot z_j\right] \cdot \left(x_i - x_j\right)^2 \\
&= \frac{M \cdot (M-1)}{N \cdot (N-1)} \cdot \left(x_i - x_j\right)^2,
\end{aligned}$$

and so it follows that

$$\begin{aligned}\mathbf{E}\left[\sum_{i\in[N]}\sum_{j\in[N]} z_i\cdot z_j\cdot(x_i-x_j)^2\right] &= \frac{M\cdot(M-1)}{N\cdot(N-1)}\cdot\sum_{i\in[N]}\sum_{j\in[N]}(x_i-x_j)^2\\ &= 2\cdot M\cdot(M-1)\cdot\sigma^2.\end{aligned}$$

In particular, defining

$$\begin{aligned}\widehat{\sigma^2} &= \frac{1}{2\cdot M\cdot(M-1)}\sum_{i\in[N]}\sum_{j\in[N]} z_i\cdot z_j\cdot(x_i-x_j)^2\\ &= \frac{1}{M-1}\cdot\sum_{i\in[N]} z_i\cdot(x_i-\hat{\mu})^2,\end{aligned}$$

we have an unbiased estimator of σ^2, and hence can derive an unbiased estimator of $\mathbf{Var}(\hat{\mu})$.

Algorithm

As such, we can compute as follows:

1. Sample $\mathcal{S}\sim\mathsf{Unif}\left(\binom{[N]}{M}\right)$.
2. For $i\in\mathcal{S}$, compute x_i, and hence

$$\hat{\mu}=\frac{1}{M}\sum_{i\in\mathcal{S}}x_i,\qquad \widehat{\sigma^2}=\frac{1}{M-1}\cdot\sum_{i\in\mathcal{S}}(x_i-\hat{\mu})^2.$$

3. Compute

$$\begin{aligned}\widehat{\mathsf{se}}^2(\widehat{\mu}) &= \frac{N-M}{M\cdot N}\cdot\widehat{\sigma^2}\\ &= \frac{N-M}{N}\cdot\frac{1}{M}\cdot\sum_{i\in\mathcal{S}}(x_i-\hat{\mu})^2.\end{aligned}$$

4. For large (M,N), report an approximate $(1-\alpha)$ confidence set as

$$C_\alpha=\left[\hat{\mu}\pm\zeta_{\frac{\alpha}{2}}\cdot\widehat{\mathsf{se}}(\widehat{\mu})\right],$$

   where ζ denotes a suitable quantile of the standard normal distribution.